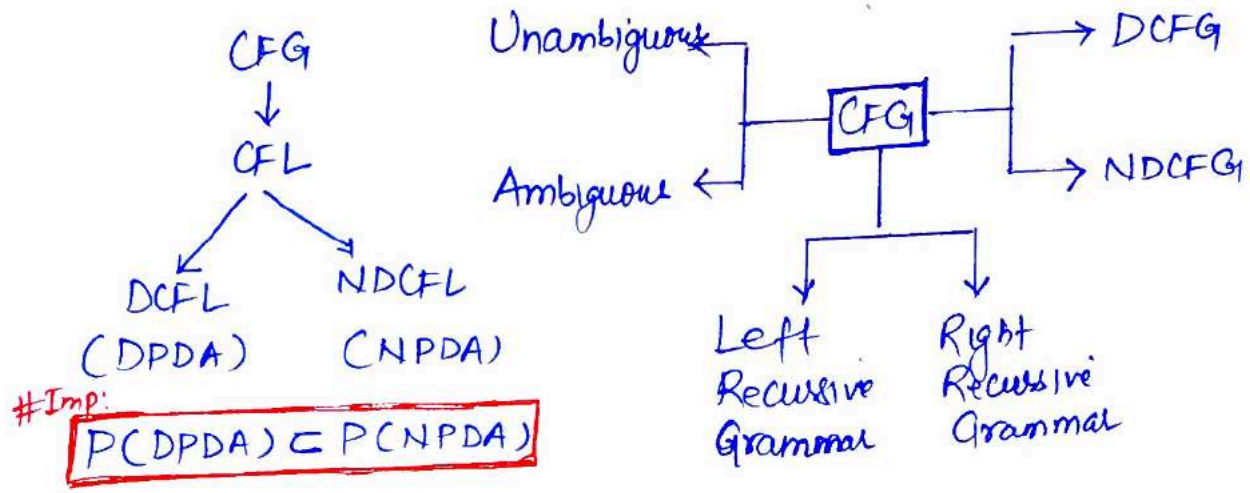
Unit-03

Context free Grammar (CFG):- A Grammar G is a CFG if every Production is of the form $A \to \beta$, where $A \in V$, $\beta \in (V \cup T)^*$

Ex: $S \to aSa | bSb | \varepsilon$



Derivation:

- It is a Process of deriving a string
- A string $\omega \in L(G)$ iff there is atleast one derivation for it

Derivation Tree OR Parse Tree: Graphical representation of derivation Process

- How to represent a derivation tree ?
  - Root node is always start symbol
  - Internal nodes are always non terminals
  - Leaf node is always terminal node

NOTE:- Derivation is read from Left to Right

Ex:- G : $S \to aAB | a$, $A \to aBA | a$, $B \to b$ Construct Parse tree for String aabab

Sentence

Sentencial form

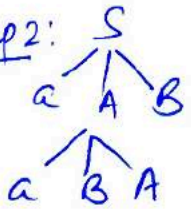
Start Symbol

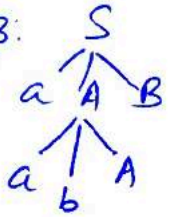
Derivation:- $S \Rightarrow aAB \Rightarrow aaBAB \Rightarrow aabAB \Rightarrow aabaB \Rightarrow aabab$

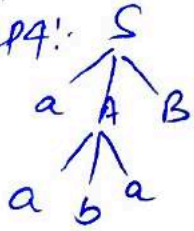
Derivation Process Sign

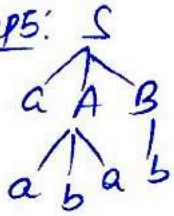
Parse tree OR Derivation tree: Graphical representation of derivation Process.

Step1: 

Step2: 

Step3: 

Step4: 

Step5: 

parse tree.

Ans.

Types of derivation Process: two types

1- Leftmost derivation (LMD):- A derivation $S \overset{*}{\Rightarrow} w$ is called a Leftmost derivation if we apply a Production only to the Leftmost variable at every step.

Ex:- Consider a Grammar G_1 for the Language $L = \{a^{2n} b^m, m, n \geq 0\}$

G_1: $S \to AB$, $A \to aaA / \varepsilon$, $B \to bB / \varepsilon$, find LMD for string $w = aab$

Leftmost derivation tree:

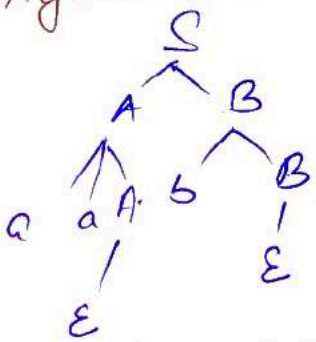
$S \Rightarrow AB$
$\Rightarrow aaAB$; $A \to aaA$
$\Rightarrow aaB$; $A \to \varepsilon$
$\Rightarrow aabB$; $B \to bB$
$\Rightarrow aab$; $B \to \varepsilon$

2- Right most derivation (RMD):- A derivation $S \overset{*}{\Rightarrow} w$ is a rightmost derivation if we apply a Production to the right most variable at every step

Ex:- RMD for string $w = aab$, Consider above grammar.

Rightmost derivation tree:

$S \Rightarrow AB$
$\Rightarrow AbB$; $B \to bB$
$\Rightarrow Ab$; $B \to \varepsilon$
$\Rightarrow aaAb$; $A \to aaA$
$\Rightarrow aab$; $A \to \varepsilon$



Ex:- $G\{S \to 0B / 1A, A \to 0 / 0S / 1AA, B \to 1 / 1S / 0BB\}$ for the String 00110101 find LMD & RMD:

LMD:

$S \Rightarrow 0B$
$\Rightarrow 00BB$; $B \to 0BB$
$\Rightarrow 001B$; $B \to 1$
$\Rightarrow 0011S$; $B \to 1S$
$\Rightarrow 00110B$; $S \to 0B$
$\Rightarrow 001101S$; $B \to 1S$
$\Rightarrow 0011010B$; $S \to 0B$
$\Rightarrow 00110101$; $B \to 1$

RMD:

$S \Rightarrow 0B$
$\Rightarrow 00BB$; $B \to 0BB$
$\Rightarrow 00B1S$; $B \to 1S$
$\Rightarrow 00B10B$; $S \to 0B$
$\Rightarrow 00B101S$; $B \to 1S$
$\Rightarrow 00B1010B$; $S \to 0B$
$\Rightarrow 00B10101$; $B \to 1$
$\Rightarrow 00110101$; $B \to 1$

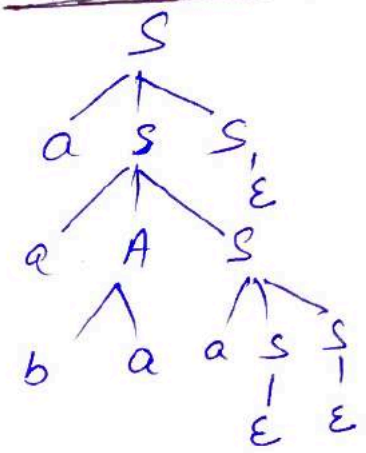
Ex:- $S \to aAS \mid ass \mid \varepsilon$, $A \to SbA \mid ba$, find Leftmost and rightmost derivation tree for string $\omega = aabaa$

Sol^n:- Given string $\omega = aabaa$.

LMD:

$S \Rightarrow aSS$
$\Rightarrow aaASS$; $S \Rightarrow aAS$
$\Rightarrow aabaSS$; $A \to ba$
$\Rightarrow aabaaSSS$; $S \to aSS$
$\Rightarrow aabaaSS$; $S \to \varepsilon$
$\Rightarrow aabaaS$; $S \to \varepsilon$
$\Rightarrow aabaa$; $S \to \varepsilon$

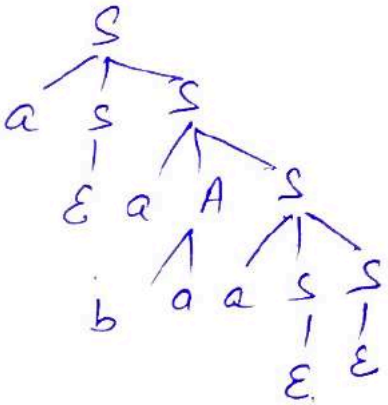
Left most derivation tree:



RMD:

$S \Rightarrow aSS$
$\Rightarrow aSaAS$; $S \to aAS$
$\Rightarrow aSaAaSS$; $S \to aSS$
$\Rightarrow aSaAaS$; $S \to \varepsilon$
$\Rightarrow aSaAa$; $S \to \varepsilon$
$\Rightarrow aSabaa$; $A \to ba$
$\Rightarrow aabaa$; $S \to \varepsilon$

Rightmost derivation Tree:



Ex:- Consider the Grammar $S \to S+S \mid S*S \mid a \mid b$ find Left most and Right most derivations for string $\omega = a*a+b$

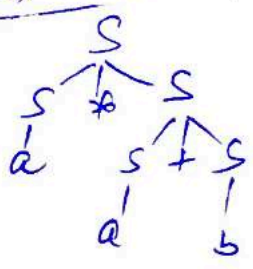
Sol^n: LMD:

$S \Rightarrow S*S$
$\Rightarrow a*S$
$\Rightarrow a*S+S$
$\Rightarrow a*a+S$
$\Rightarrow a*a+b$

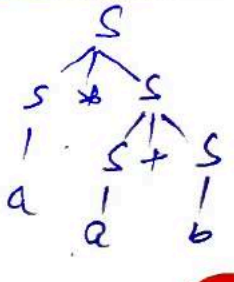
RMD:

$S \Rightarrow S*S$
$\Rightarrow S*S+S$
$\Rightarrow S*S+b$
$\Rightarrow S*a+b$
$\Rightarrow a*a+b$

Leftmost derivation Tree:



Rightmost derivation Tree:

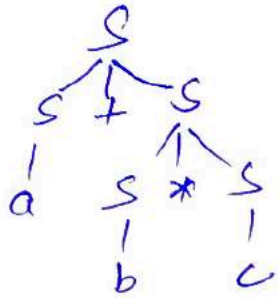


Ambiguous Grammar: A Grammar G1 is ambiguous grammar if $\exists w \in L(G)$ such that w has > 1 P.T / D.T / S.T / LMD / RMD } Either using 2LMD or 2RMD

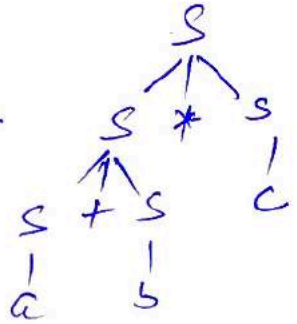
i.e a grammar G1 is ambiguous if there is more than one Parsetree or LMD/RMD for a string $w \in L(G)$

Ex:- G1 : $S \to S+S \mid S*S \mid a \mid b \mid c$, Grammar G1 is ambiguous because There exist two different LMD for a string $w = a+b*c$

LMD:



OR



NOTE:- A Grammar G1 is unambiguous if there exist exactly one Parsetree or LMD/RMD for all string $w \in L(G)$

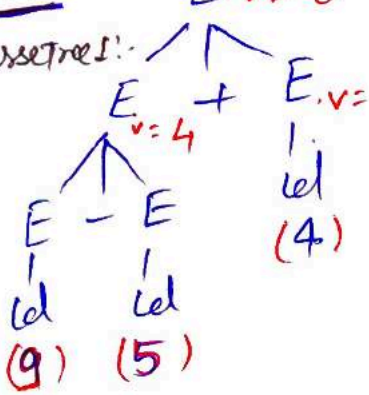
NOTE:- if a Grammar G1 is ambiguous, it doesn't mean language(L) is ambiguous

Ex:- State whether given grammar is ambiguous or not G1: $E \to E+E \mid E-E \mid id$

Soln: We need take one string $w \in L(G1)$ if there are more than one LMD/R Then given grammar is ambiguous. Also if L(G) has more than one Parsetree for w (Let $w = 9-5+4$) Then given grammar is ambiguous

LMD:- $E.v = 8$

Parsetree 1:



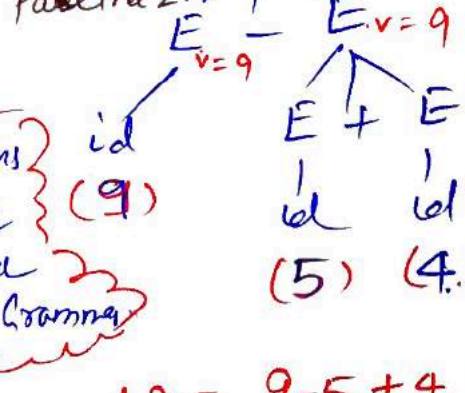
$w = 9-5+4 = 4$

OR

Two different Ans for 9-5+4 b/c given Grammar is Ambiguous Gramma

LMD:- $E.v = 9-9 = 0$

Parsetree 2:

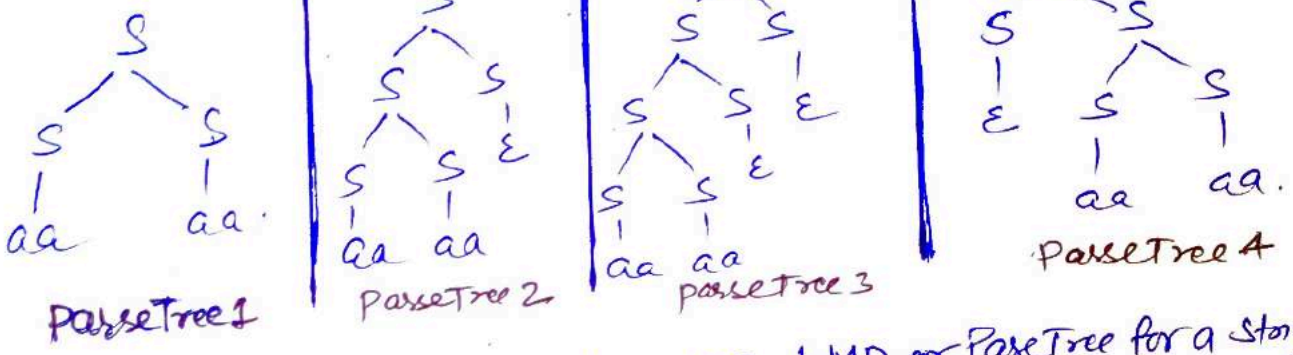


$w = 9-5+4 = 0$

Ex:- State whether given grammar is ambiguous or not

G: $S \to aa|bb|SS|\varepsilon$

Soln: We need take one string $w \in L(G)$ if there are more than one LMD/RMD then given grammar is ambiguous. Also if $L(G)$ has more than one Parse tree for w (Let $w = aaaa$) then given grammar is ambiguous



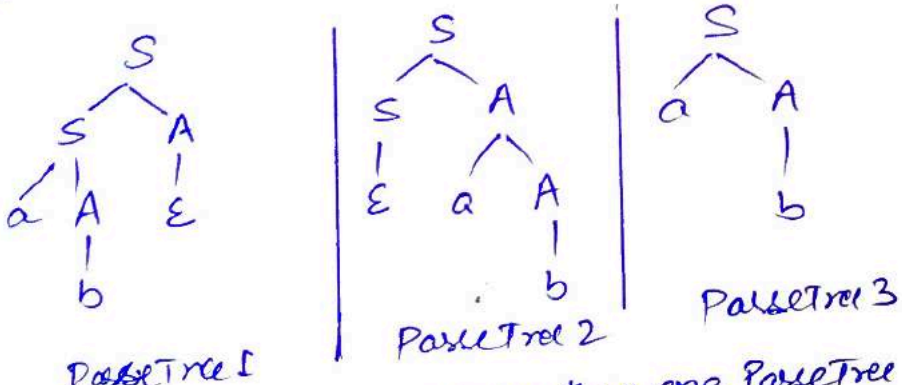
We see that there are more than one LMD or ParseTree for a string $w \in L(G)$

Therefore, given grammar is ambiguous.

Ex:- State whether given grammar is ambiguous or not

G: $S \to SA|aA|\varepsilon$; $A \to aA|b|\varepsilon$

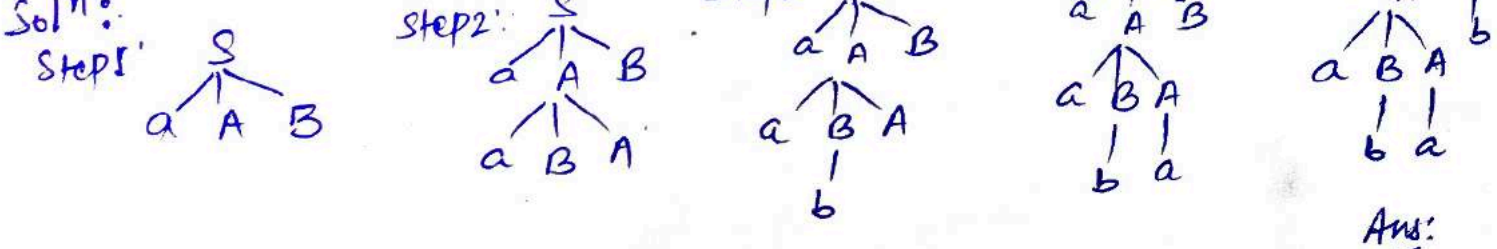
Soln: Let $w = ab$ if there are more than 1 parsetree then G is ambiguous



We see that there are more than one ParseTree for a string $w \in L(G)$. Therefore given grammar is ambiguous.

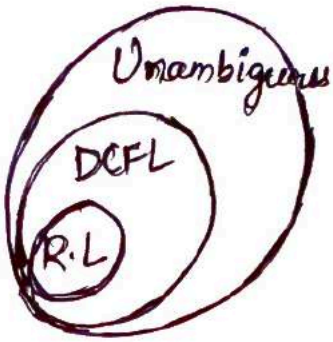
Ex:- find Parsetree for the string $w = aabab$ such that $w \in L(G)$, G: $S \to aAB|a$; $A \to aBA|a$; $B \to b$

Soln:

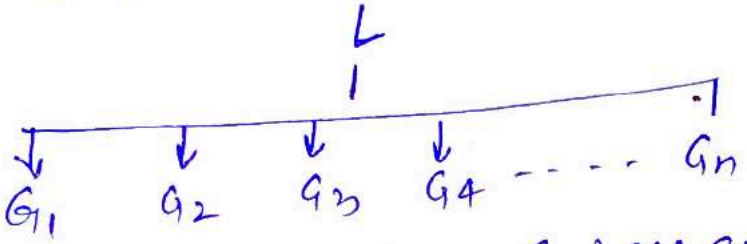


NOTE:-

- Regular grammar → Ambiguous (NFA), Unambiguous (DFA)
- Regular language always Unambiguous
- DCFL is always Unambiguous language.
- Ambiguity start from CFL



Inherent ambiguous language :- if all grammar ambiguous for a language(L) Then the language (L) is Inherently ambiguous language



- if all grammar ($G_1, G_2, G_3, G_4 \cdots G_n$) are ambiguous Then The language L is Inherently ambiguous language
- if any one grammar ($G_1, G_2, G_3, G_4 \cdots G_n$) is Unambiguous Then The language L is Unambiguous language.
- Inherent word used for language not for Grammar.

i.e Grammar < Ambiguous, Unambiguous

language < Inherently ambiguous, Unambiguous.

Ex:- Check wether the given language $L = \{a^n / n \geq 0\}$ is a Inherently ambiguous or not

Sol^n:

$L = \{a^n / n \geq 0\}$ ← Unambiguous language.

$G_1 = \{S \to aS/a/\varepsilon\}$ Ambiguous Grammar

$G_2 = \{S \to Sa/a/\varepsilon\}$ Ambiguous Grammar

$G_3 = \{S \to aS/\varepsilon\}$ UnAmbiguous Grammar

$G_n = \{$ . . .

G_3 is UnAmbiguous Grammar for L Therefore L is not Inherently Ambiguous

Ex:- $L = \{a^m b^n c^k / m, n, k \geq 1$, either $m = n$ or $n = k\}$

Sol^n:- Case 1: if $m = n$ Then $L = \{a^m b^m c^k / m, k \geq 1\} = L_1$

Case 2: if $n = k$ Then $L = \{a^m b^n c^n / m, n \geq 1\} = L_2$

Now: $L_1 \cap L_2 = \{a^m b^m c^m / m \geq 1\}$, Therefore no Unambiguous Grammar for L, all are Ambiguous Grammar, Hence L is Inherently ambiguous language.

Remove Ambiguity:

- Causes Such as Left recursion, Common Prefixes etc. makes the grammar ambiguous.
- The removal of these causes may Convert the grammar into Unambiguous Grammar.
- However, it is not always Compulsory.

NOTE:- It is not always Possible to Convert an ambiguous grammar into an Unambiguous grammar b/c Ambiguity finding & Removal both are Undecidable

Removing Ambiguity by Precedence & Associativity rules:

An ambiguous Grammar may be converted into an Unambiguous grammar by implementing -
- Precedence Constraints
- Associativity Constraints

These Constraints are implemented using the following rules:

Rule-01:- The Precedence Constraint is implemented using the following rules
- The level at which the Production is Present defines the Priority of the operator Contained in it.
- The higher the level of the Production, the lower the Priority of operator.
- The lower the level of the Production, the higher the Priority of operator.

Rule-02:- The Associativity Constraint is implemented using the following rules
- if the operator is Left associative, induce Left recursion in its Production.
- if the operator is Right associative, induce right recursion in its Production.

Ex:- $G_1: E \to E+E \mid E*E \mid id$. Ambiguous Grammar Convert into Unambiguous Gram

Soln: $G_1: E \to E+E$, $E \to E*E$, $E \to id$ (Ambiguous Grammar) $\xrightarrow[\text{Apply Rule 1 \& 2}]{\text{UnAmbiguous G.}}$ $G_1: E \to E+T \mid T$, $T \to T*F \mid F$, $F \to id$ (UnAmbiguous Grammar) — Ans

Now precedence: $id > * > +$; $+ > +$ and $* > *$ } b/c Left Associative

Ex: Convert the following Ambiguous Grammar into Unambiguous Grammar.
$G: R \to R+R \mid R.R \mid R* \mid a \mid b$

Soln: $G: R \to R+R$, $R \to R.R$, $R \to R^*$, $R \to a \mid b$ | using the Precedence and Associativity rules, we write The Corresponding Unambiguous grammar as $\Rightarrow$

$E \to E+T \mid T$
$T \to T.F \mid F$
$F \to F* \mid G$
$G \to a \mid b$

OR

$E \to E+T \mid T$
$T \to T.F \mid F$
$F \to F* \mid a \mid b$

Ans

Ex:- Convert the following ambiguous grammar into Unambiguous grammar.

bexp → bexp OR bexp / bexp and bexp / not bexp / T / F, where bexp represents Boolean expression, T represents TRUE and F represents FALSE.

Sol^n: To convert the given grammar into its corresponding Unambiguous grammar, we implement the Precedence and associativity Constraints.

The Priority order is (T, F) > not > and > or

where
- and operator is Left associative
- or operator is left associative

bexp → bexp OR bexp
/ bexp and bexp
/ not bexp
/ TRUE
/ FALSE

Unambiguous
Apply Rules 1 & 2

E → E or F / F
F → F and G / G
G → Not G / T / F

Ans Un Ambiguous Grammar

Ex: Ambiguous Grammar.

G { E → E+E
E → E*E
E → E↑E
E → id

Remove Ambiguity
Apply Rules 1 & 2.

E → E + T / T (+ Left Associative)
T → T * F / F (* Left Associative)
F → G ↑ F / G (↑ Right Associative)
G → id

The priority order: id > ↑ > * > +

Associativity: + and * Left Associative and ↑ operator Right Associative

Ex: Find precedence & Associativity

(i) G_1: { A → A $ B / B
B → B # C / C
C → D @ C / D
D → d

According to Given Grammar.

Sol^n. (i) @ > # > $

$: Left Associative
: Left Associative
@ : Right Associative

(ii) G_2: { E → E * F / F + E / F
F → F − F / id

According to Given Grammar.

(iii) [* = +] > −

*: Left Associative
+: Right Associative
−: Both Left & Right Associative

SIMPLIFICATION OF CFG

- In CFG, it may happen that all the production rules and symbols are not needed for the derivation of strings.
  Elimination of these productions & symbols is called **simplification of CFG**
- It consists of three steps
  1. Removal of Useless Production
  2. Removal of Unit Production
  3. Removal of NULL Production.

REMOVAL OF USELESS PRODUCTION

- Variables/Non-Terminals which don't derive any string are called as useless symbols.
- The production rule generating useless symbol becomes useless production.
- eg 1: { S → AB|a ; A → BC|b ; B → eD|e }

  In above grammar useless productions are

  A → BC

  B → eD

  Because Variable 'C' & 'D' are not deriving any string. Hence after removal of useless product^{n} We have grammar as

  { S → AB|a ; A → b ; B → e }

eg 2: L(G) ; { S → AB ; A → aA|bC ; B → bB|a ; D → e }

In above grammar useless productions are

A → bC

D → e.

Variable C is not deriving any string.
Variable D is not generated in any of the product^{n}
We have grammar after removing useless product^{n} as

{ S → AB ; A → aA ; B → bB|a }

Removal of Unit Production

- Any rule in the form <Variable> → <Variable> i.e. $X \rightarrow Y$ is called unit product
- Substitute all Y product" if Y is generating any terminal symbol.
- e.g. 1. $\{S \rightarrow A \ ; \ A \rightarrow a|b\}$

  Substitute variable A in the production $S \rightarrow A$ with the terminal 'a' & 'b' because $A \rightarrow a \ ; A \rightarrow b$
  After removing unit production we have
  $\{S \rightarrow a|b\}$

e.g. 2. $\{S \rightarrow AB \ ; \ A \rightarrow a|b \ ; \ B \rightarrow A\}$

Rule $B \rightarrow A$ is the unit production.
We can substitution A with 'a' & 'b'
∴ After removing unit production. We have,

$\{S \rightarrow AB \ ; \ A \rightarrow a|b \ ; \ B \rightarrow a|b\}$

e.g. 3 $\{S \rightarrow A|bb \ ; \ A \rightarrow B|b \ ; \ B \rightarrow S|a\}$

Unit productions in above grammar are

$S \rightarrow A$	; substitute	A with	b & B	$S \rightarrow b \mid B$
$A \rightarrow B$	; "	B "	a & S	$A \rightarrow a \mid S$
$B \rightarrow S$	; "	S "	bb & A	$B \rightarrow bb \mid A$

After substituting values we again get unit production

$S \rightarrow B$	; substitute	B with	a & S	$S \rightarrow a \mid S$
$A \rightarrow S$	; "	S "	bb & A	$A \rightarrow bb \mid A$
$B \rightarrow A$	; "	A "	b & B	$B \rightarrow b \mid B$

Again we get unit production

$S \rightarrow S$	; substitute	S with	bb & A	$S \rightarrow bb \mid A$
$A \rightarrow A$	; "	A "	b & B	$A \rightarrow b \mid B$
$B \rightarrow B$	; "	B "	a & S	$B \rightarrow a \mid S$

Grammar we get

$\{S \rightarrow a|b|bb \ ; \ A \rightarrow a|b|bb \ ; \ B \rightarrow a|b|bb\}$

Removal Of Null Production

- Any production of the form $X \rightarrow \varepsilon$ should be removed. (X → variable)
- Removal Procedure
  - ↳ find all variables which derive ε. eg $X \rightarrow \varepsilon$
  - ↳ If variable X is present in any production then remove X from the production.
  - ↳ Combine original productions with above step & remove ε productions.

eg 1. $L(G) : \{ S \rightarrow aA|bBA|AB \; ; \; A \rightarrow aA|\varepsilon \; ; \; B \rightarrow b \}$

There is one NULL production in above grammar i.e $A \rightarrow \varepsilon$

Variable A is present in 4 production rules i.e ($S \rightarrow aA|bBA|AB$; $A \rightarrow aA$). Remove A from the product

$S \rightarrow a|bB|B$; $A \rightarrow a$

Now combine original productions with above step & remove ε productions. We have

$L(G') : \{ S \rightarrow aA|bBA|AB|a|bB|B \; ; \; A \rightarrow aA|a \; ; \; B \rightarrow b \}$

Ans

eg 2. $L(G) : \{ S \rightarrow ASA|aB|b \; ; \; A \rightarrow B|\varepsilon \; ; \; B \rightarrow b|\varepsilon \}$

There are two NULL productions in above grammar

$A \rightarrow \varepsilon$ & $B \rightarrow \varepsilon$

Variable A is present in one production rule i.e $S \rightarrow ASA$
Variable B is present in two production rules i.e $S \rightarrow aB$; $A \rightarrow B$

After removing A & B from product respectively we get-

$S \rightarrow AS|SA|S$; $S \rightarrow a$; $A \rightarrow b$

Now combine original productions with above step & remove ε production. we have

$L(G') : \{ S \rightarrow ASA|aB|b|AS|SA|S|a \; ; \; A \rightarrow B|b \; ; \; B \rightarrow b \}$

Ans

CHOMSKY NORMAL FORM (CNF)

— A CFG is in CNF if the productions are in following forms

$\langle Variable \rangle \rightarrow \langle Terminal \rangle$

$\langle variable \rangle \rightarrow \langle variable \rangle \langle variable \rangle$

$S \rightarrow \varepsilon$ { start symbol can generate ε }

ex1: $A \rightarrow a$
$A \rightarrow AB$

ex2: $S \rightarrow AB \mid \varepsilon$
$A \rightarrow a$
$B \rightarrow b$

ex3: $A \rightarrow BC \mid a$
$B \rightarrow CD \mid b$
$C \rightarrow e$
$D \rightarrow d$

ex4: $S \rightarrow BA \mid a$
$B \rightarrow a$
$A \rightarrow b$

ex5: $S \rightarrow AB \mid BD$
$A \rightarrow CD \mid a \mid AC$
$C \rightarrow DE \mid e$
$B \rightarrow b$
$D \rightarrow d$
$E \rightarrow f$

— ALGO TO CONVERT CFG into CNF

① If start symbol occurs on R.H.S of production rule, create a new start symbol S' & a new production $S' \rightarrow S$

② Remove NULL production

③ Remove UNIT production

④ If R.H.S of production contains more than two variable then two consecutive variables can be replaced by a new variable.

ex: $A \rightarrow XYZ \Rightarrow A \rightarrow XP$, $P \rightarrow YZ$ or $A \rightarrow PZ$, $P \rightarrow XY$

Here, we can replace either YZ or XY with new variable

⑤ If R.H.S of production contains combination of Terminal & Variable then each terminal symbol is replaced by new variable.

ex. i) $A \rightarrow aB \Rightarrow A \rightarrow XB$ (CNF form), $X \rightarrow a$ "

ii) $A \rightarrow BCDE \Rightarrow A \rightarrow BX$ (CNF form), $X \rightarrow CY$ ", $Y \rightarrow DE$ "

{ Here also we can take any consecutive two variables & replace with with new var.

Q1. Construct equivalent CNF for given CFG

a) $S \rightarrow aSa \mid bSb \mid a \mid b$

Sol^n: Since S appears in R.H.S, we add new production, we have

$S' \rightarrow S$ (Not in CNF)
$S \rightarrow aSa$ "
$S \rightarrow bSb$ "
$S \rightarrow a$ (CNF)
$S \rightarrow b$ (CNF)

NO NULL/UNIT production.

We find a production $S \rightarrow aSa$ & $S \rightarrow bSb$ where R.H.S contains combination of variable & terminal. Apply step (5)

We have $S \rightarrow AX$, $X \rightarrow SA$, $A \rightarrow a$ & $S \rightarrow BY$, $Y \rightarrow SB$, $B \rightarrow b$

final production set we obtain is as follows

$\{ S' \rightarrow AX \mid BY$; $S \rightarrow AX \mid BY \mid a \mid b$
$X \rightarrow SA$
$Y \rightarrow SB$
$A \rightarrow a$
$B \rightarrow b$
$\}$

b.) $S \rightarrow ASA \mid aB$; $A \rightarrow B \mid S$; $B \rightarrow b \mid \varepsilon$

Sol^n: We add new production $S' \rightarrow S$ because S appears in R.H.S

$S' \rightarrow S$
$S \rightarrow ASA$ (Not in CNF)
$S \rightarrow aB$ "
$A \rightarrow B$ (UNIT production, Not in CNF)
$A \rightarrow S$ ("
$B \rightarrow b$ (CNF)
$B \rightarrow \varepsilon$ (NULL production, Not in CNF)

After Removing NULL production $B \rightarrow \varepsilon$, the production set becomes

$S' \rightarrow S$
$S \rightarrow ASA \mid AS \mid SA \mid S$
$S \rightarrow aB \mid a$
$A \rightarrow B \mid S$
$B \rightarrow b$

After removing $B \rightarrow \varepsilon$, we have
$S' \rightarrow S$; $S \rightarrow ASA \mid aB \mid a$; $A \rightarrow B \mid S \mid \varepsilon$; $B \rightarrow b$

Now, Removing $A \rightarrow \varepsilon$, we have set of production as

Now remove unit productions

After removing $S \to S$, the product" set becomes

$S' \to S$; $S \to ASA|AS|SA|aB|a$; $A \to B|S$; $B \to b$

After removing $A \to B$, the product" set becomes

$S' \to S$; $S \to ASA|AS|SA|aB|a$; $A \to S|b$; $B \to b$

After removing $S' \to S$ & $A \to S$, the product" set becomes

$S' \to ASA|aB|a|AS|SA$

$S \to ASA|aB|a|AS|SA$

$A \to ASA|aB|a|AS|SA|b$

$B \to b$

Now find more than two variables in the product"

Here $S' \to ASA$, $S \to ASA$ & $A \to ASA$ violates CNF form.

Using step (4) we have production set

$S' \to AX|aB|a|AS|SA$

$S \to AX|aB|a|AS|SA$

$A \to AX|aB|a|AS|SA|b$

$B \to b$

$X \to SA$

Using set (5) fix the problem in R.H.S for the product" containing combination of variable & terminal.

final production set becomes

$\{ S' \to AX|YB|a|AS|SA$

$S \to AX|YB|a|AS|SA$

$A \to AX|YB|a|AS|SA|b$

$B \to b$

$X \to SA$

$Y \to a$

$\}$

Ans.

c) $S \to bA \mid aB \;;\; A \to bAA \mid aS \mid a \;;\; B \to aBB \mid bS \mid b$

Sol^n = Create New start symbol S' & new product^n $S' \to S$. We have

$S' \to S$ (UNIT product^n)
$S \to bA$ (Not in CNF)
$S \to aB$ "
$A \to bAA$ "
$A \to aS$ "
$A \to a$ (CNF)
$B \to aBB$ (Not in CNF)
$B \to bS$ "
$B \to b$ (CNF)

There is no NULL/UNIT production^n except $S' \to S$ & more than two variables After removing combination var & Terminals, We have final set of production rules

$S' \to XA \mid YB$
$S \to XA$
$S \to YB$
$A \to ZA$
$A \to YS$
$A \to a$
$B \to WB$
$B \to XS$
$B \to b$
$X \to b$
$Y \to a$
$Z \to XA$
$W \to YB$

No: of productions (max^m)
$= (K-1) + |P| + |T|$
$= (5-1) + 8 + 2$
$= 4 + 8 + 2$
$= 14$

Gate exam.

Note: Let G be CFG without NULL production & UNIT product^n 'K' be the max^m no: of symbols on R.H.S of any product^n, then Equivalent CNF contains max^m no: of product^n

$= (K-1) + |P| + |T|$

(K-1) → no: of symbols; |P| → production rules; |T| → Terminal

GREIBACH NORMAL FORM (GNF)

- A CFG is in GNF if the productions are in the following forms

  <variable> → <Terminal>

  <variable> → <single Terminal> <variable$>

  S → ε

  Start symbol

ex 1: A → a
B → aBB | a

ex 2: A → aABBA | b
B → a

ex 3: S → aAABBB
A → a
B → b

ex 4: S → bAAAABBB
A → a
B → b

- ALGO TO CONVERT CFG INTO GNF

1. If start symbol S occurs in any of R.H.S rule, create a new start symbol S' & add new production S' → S.
2. Remove Null productions
3. Remove Unit productions
4. Remove all direct and indirect **left-recursion**
5. Do proper substitutions of productions
   a) Convert CFG to CNF
   b) Rename all variables as $A_1, A_2, A_3 \ldots\ldots A_n$
   c) For every production A_i do following
      (i) If $A_i \rightarrow A_j X$; $i<j$ then apply substitution of A_j
      (ii) If $A_K \rightarrow A_K X$; then Remove left recursion

      ] include this if step 4 not included in algo

Q1. Convert given CFG into equivalent GNF

a) $S \rightarrow XY \mid Xn \mid P$

$X \rightarrow mX \mid m$

$Y \rightarrow Xn \mid O$

Sol^n: Here S doesn't appear in any of R.H.S product.
There are no NULL/UNIT production.
Using step ⑤ a) convert CFG to CNF
We have set of production

$S \rightarrow XY \mid XN \mid P$

$X \rightarrow mX \mid m$

$Y \rightarrow XN \mid O$

$N \rightarrow n$

Apply step ⑤ c) Apply substitution after renaming all var.

$A_1 \rightarrow A_2A_3 \mid A_2A_4 \mid P$

$A_2 \rightarrow mA_2 \mid m$

$A_3 \rightarrow A_2A_4 \mid O$

$A_4 \rightarrow n$

} Rename variables

In $A_1 \rightarrow A_2A_3 \mid A_2A_4$ we can substitute value of A_2. Also in $A_3 \rightarrow A_2A_4$ we can substitute $A_2 \rightarrow mA_2 \mid m$.
final set of production is as follows.

$A_1 \rightarrow mA_2A_3 \mid mA_3 \mid mA_2A_4 \mid mA_4 \mid P$

$A_2 \rightarrow mA_2 \mid m$

$A_3 \rightarrow mA_2A_4 \mid mA_4 \mid O$

$A_4 \rightarrow n$

Ans

b.) $B \to AB|AC|DA$; $A \to a$; $C \to DA$; $D \to d$

Sol^n Given CFG has start symbol 'B' which is present in the R.H.S of the production also. Therefore create new start symbol B' & new production $B' \to B$. The set of production we have is

$B' \to B$; $B \to AB|AC|DA$; $A \to a$; $C \to DA$; $D \to d$

There are no NULL/UNIT production.
There is no left recursion in the given grammar
Convert CFG into CNF. We have set of production as

$B' \to AB|AC|DA$; $B \to AB|AC|DA$; $A \to a$; $C \to DA$; $D \to d$

Now substitute all variables with new name

$A_1 \to A_2A_3|A_2A_4|A_5A_2$; $A_3 \to A_2A_3|A_2A_4|A_5A_2$; $A_2 \to a$; $A_4 \to A_5A_2$; $A_5 \to d$

Apply substitution using step ③ c.)(i)

$A_1 \to aA_3|aA_4|dA_2$

$A_3 \to aA_3|aA_4|dA_2$

$A_2 \to a$

$A_4 \to dA_2$

$A_5 \to d$

Ans.

Elimination of Left Recursion

If production is of form $A \to AX|t$ that means left recursion is present.

(A: variable, t: Terminal, A: same symbol)

Recursion is present due to Variable A because variable in L.H.S is equal to first variable in R.H.S

To remove left recursion create new symbol variable A' & new production like below

A->tA' | t (Replace A with t, X with new variable A' | t is a terminal)

A'->XA'|X (New production for new variable A')

NOTE: variable X must derive a terminal, otherwise left recursion will be there in new production too.